



Q.1. Solve the following:

(6x5=30)

- Define $T : R^3 \rightarrow R^3$ by $T(x_1, x_2, x_3) = (-x_3, x_1, x_1 + x_3)$. Find $N(T)$. Is T one-to-one?
- Check whether the first quadrant $W = \{(x, y) : x, y \geq 0\}$ makes a subspace of R^2 or not.
- Find the reduced echelon form of the matrix
$$\begin{bmatrix} 1 & -2 & 3 & -1 \\ 2 & -1 & 2 & 2 \\ 3 & 1 & 2 & 3 \end{bmatrix}$$
- Show that Similar matrices having same eigenvalues.
- Show that if $u \in V$ is orthogonal to every $v \in V$, then $u = 0$.
- Give an example of non-similar matrices having same characteristic polynomial.

Q.2. Solve the following:

(5x6=30)

- Determine the values of a for which the system of linear equations has no solution, exactly one solution and infinitely many solutions.
$$\begin{aligned} x + y + 7z &= -7 \\ 2x + 3y + 17z &= -16 \\ x + 2y + (a^2 + 1)z &= 3a. \end{aligned}$$
- If possible, find the inverse of the matrix
$$\begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{bmatrix}$$
- Find Eigen Values and Eigen vectors of
$$\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$
- Find solution of the given system by Gauss-Jordan Elimination method.
$$2x_1 - x_2 - x_3 = 4 ; 3x_1 - 2x_3 + 4x_2 = 11 ; 3x_1 - 2x_2 + 4x_3 = 11$$
- Find an equation of the subspace W of R^3 spanned by $v_1 = (1, -3, 5)$, $v_2 = (-2, 6, -10)$.